

Ebola Virus and Marburg Virus Infections

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Marburg virus and Ebola virus infections cause bleeding and organ malfunction. These infections often result in death.

- Marburg and Ebola infections are spread through handling live or dead infected animals or by contact with skin or body fluids of an infected person who has symptoms or who has died.
- Symptoms begin with a fever, muscle and body aches, headache, cough, and a sore throat, followed by vomiting and diarrhea as well as bleeding from the mouth, nose, or internal organs.
- To confirm the diagnosis, doctors do blood and urine tests.
- Treatment includes giving fluids by vein and other treatments to maintain body functions.
- Strict isolation and sanitary precautions (and, for health care professionals, use of protective clothing) are needed to prevent spread of the infection.

Marburg and Ebola virus infections are [hemorrhagic fevers](#), characterized by bleeding. They are caused by viruses called [filoviruses](#). Ebola virus has 5 species including Zaire Ebola virus and Sudan Ebola virus.

Outbreaks of Ebola and Marburg virus infections started in Central and West Africa, probably because the infections are common in wild animals there. Such outbreaks are rare and, in the past, have not spread very far because they typically occurred in isolated, rural areas. However, travelers from those areas sometimes carry the infection to nearby areas or, rarely, outside of Africa.

Transmission of Ebola and Marburg viruses

The Ebola and Marburg viruses are highly contagious.

These viruses usually infect animals and only occasionally spread to people. Infection in humans may occur by handling or eating an infected animal, such as a fruit bat or primate (apes and monkeys). Marburg virus infections have occurred in people exposed to bats (for example, in mines or caves).

Once a person has been infected, the infection can then spread from person to person when people come in direct contact with the skin or body fluids (saliva, blood, vomit, urine, stool, sweat, breast milk, or semen) of an infected person. People cannot transmit the infection before they develop symptoms. However, body fluids from dead people may transmit the infection, particularly to people who transport and prepare corpses for burial.

Ebola is not thought to spread through the air (for example, in droplets dispersed when people cough or sneeze).

In some people who recover, the Ebola virus is eventually completely eliminated from their system, and they are no longer contagious. However, in others, the virus may remain for a time in areas of the body that are protected from attack by the [immune system](#) (such as the eyes or semen). The genetic material of the Ebola virus can remain in semen for up to 3 months after men no longer have symptoms.

The Marburg virus can remain in semen for up to 7 weeks after men no longer have symptoms.

Symptoms of Ebola and Marburg Virus Infections

People develop the first symptoms about 2 to 21 days after exposure to the virus. At first, the symptoms usually resemble those of other, less dangerous viral infections. People have muscle aches, headache, sore throat, and cough. People also become sensitive to light, and the whites of the eyes may turn red.

Within a few days, more serious symptoms develop, such as the following:

- Abdominal pain
- Yellow skin and yellowing of the whites of the eyes ([jaundice](#))
- Severe vomiting and diarrhea

- A tendency to bleed
- Confusion, delirium, and coma

The vomiting and diarrhea cause [dehydration](#), which, if untreated leads to low blood pressure ([shock](#)), [abnormal heart rhythms](#), and death.

Bleeding begins within the first few days. Bleeding under the skin can be seen as purplish or dark spots or patches (bruises). The gums, nose, rectum, or internal organs may bleed, as may puncture wounds (as result from drawing blood or starting an intravenous line).

After about 5 days, a rash, mainly on the trunk, appears.

During the second week of symptoms, the fever breaks and either people begin to recover or their organs malfunction (called organ failure). From 25 to 90% of infected people die depending in large part on the level of medical care available. Compared with Marburg virus infection, Ebola is more likely to be fatal. Recovery takes a long time.

Diagnosis of Ebola and Marburg Virus Infections

- Blood and urine tests

Identifying Ebola in people who have been infected for only a few days is difficult because early symptoms, such as fever, can be caused by more common infections. However, if people with these symptoms are in or have traveled from areas where Ebola is occurring, doctors suspect Ebola. The Centers for Disease Control and Prevention (CDC) has provided guidelines for health care professionals to enable them to evaluate travelers returning from such areas.

If doctors suspect Ebola or Marburg virus infection, they take samples of blood and urine from the person and test them to confirm the infection.

Public health officials are notified if an Ebola or a Marburg virus infection is diagnosed.

Treatment of Ebola and Marburg Virus Infections

- Supportive care
- Antiviral medications

Supportive treatment involves:

- Giving fluids intravenously (IV) to maintain blood pressure and the balance of [electrolytes](#) (such as sodium, potassium, and calcium)
- Transfusing [clotting factors](#) to help slow or stop the bleeding
- Treating symptoms, including giving pain relievers

Two monoclonal antibody treatments (called REGN-EB3 and mAb-114) are available to treat Ebola virus infection caused by the Zaire Ebola virus. Both of these treatments were tested during the 2018 to 2020 Ebola outbreak in the Democratic Republic of the Congo and were effective, curing about 90% of people when the medication was given within the first few days after infection.

There is no effective treatment for Marburg virus infection.

Prevention of Ebola and Marburg Virus Infections

Several vaccines and antiviral medications have been developed. A Zaire [Ebola vaccine](#) is available for use in people 18 years of age and older who may be exposed to the virus (for example, certain health care professionals and scientists). Two Zaire Ebola vaccines are in routine use in the Democratic Republic of the Congo (DRC). These vaccines are only for the Zaire Ebola and do not provide cross-protection against the Sudan Ebola virus disease. Vaccines for Sudan Ebola viruses disease are being developed.

Strict isolation is needed to prevent spread. Isolating people with symptoms of possible Ebola or Marburg virus infection in facilities used only for that purpose can help. These facilities can safely handle the infected body fluids. Individuals who have been in contact with infected people should be vaccinated.

Staff members caring for people with Ebola must be completely covered in protective clothing, including head gear equipped with a respirator. Even though the infections are not thought to be spread through the air, precautions against such spread are taken. Other staff members must be available to help those wearing protective clothing remove the protective clothing.

Diligently cleaning and disinfecting surfaces and equipment that may be contaminated is crucial, as is making sure that body fluids (such as blood, sweat, vomit, feces, and urine) are handled safely.

For people who have traveled to or are in an area affected by an Ebola outbreak, the following is recommended:

- Use good hygiene, such as washing hands with soap and water or an alcohol-based hand sanitizer and avoiding contact with other people's blood and body fluids
- Avoid handling items that may have come in contact with an infected person's blood or body fluids (such as clothes, bedding, needles, and medical equipment)
- Avoid participating in funeral or burial rituals that require handling the body of someone who has died of Ebola
- Avoid contact with bats and primates (such as apes and monkeys) and do not eat raw or inadequately cooked meat prepared from these animals
- Avoid facilities in West Africa where Ebola patients are being treated
- After returning, monitor health for 21 days and seek medical care immediately if symptoms suggesting Ebola develop

Although no infections are known to have been transmitted by sexual intercourse or breastfeeding, the number of people studied has been low. Thus, there are no firm guidelines on how long people should abstain from unprotected sexual intercourse, breastfeeding, or close personal contact. However, to be safe, doctors from the World Health Organization (WHO) recommend that men refrain from sexual intercourse until their semen has been tested twice and is negative for the virus, or practice safe sex (including condom use) for 12 months from symptom onset.

